

TARalytics — Investigation Evidence

Log: logs/00000011.BIN

Generated: 2026-06-04T12:09:32

Aircraft: — · ArduCopter V4.6.3 · QUAD/X

Verification: NOT_LOADED

Assessment: MARGINAL (59/100)

Snapshots: 2

Conclusion

Verdict: MARGINAL (59/100)

Marginal flight — review the findings.

Drivers: tracking 22 · saturation

Category	Score	Grade
Overall	62	D
Attitude tracking	22	F
Control smoothness	90	A
Yaw discipline	55	F
Landing quality	80	B

Flight: 3 flight(s), 298 s armed.

Findings (11)

1. [ERROR] ERR — ERR anomaly

Subsys=12 ECode=1 @ 1552.3 s

Supporting evidence: ERR

2. [ERROR] ERR — ERR anomaly

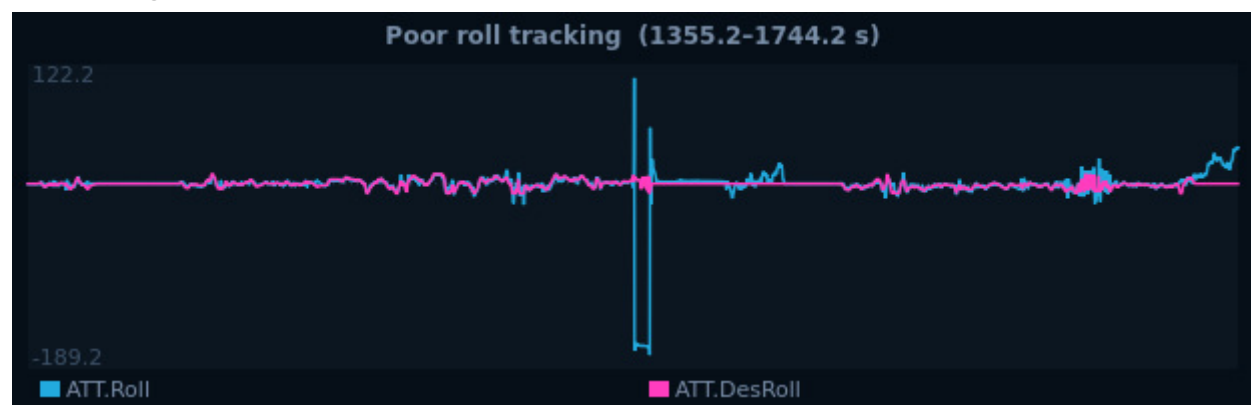
Subsys=25 ECode=1 @ 1725.6 s

Supporting evidence: ERR

3. [WARNING] TRACKING — Poor roll tracking

Roll demand-vs-response RMS 15.5°, only 93% in tolerance

Supporting evidence: ATT.Roll, ATT.DesRoll



4. [WARNING] TRACKING — Poor pitch tracking

Pitch demand-vs-response RMS 9.4°, only 91% in tolerance

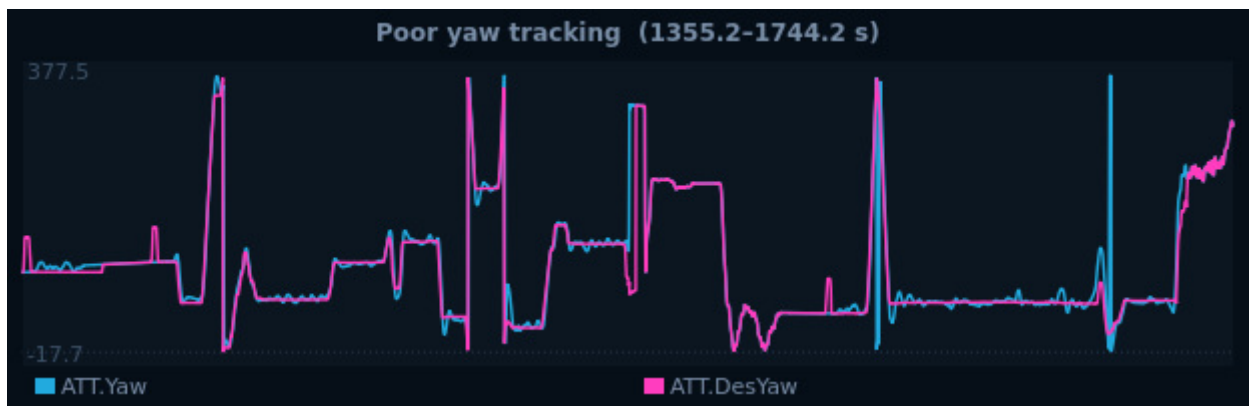
Supporting evidence: ATT.Pitch, ATT.DesPitch



5. [WARNING] TRACKING — Poor yaw tracking

Yaw demand-vs-response RMS 16.9°, only 75% in tolerance

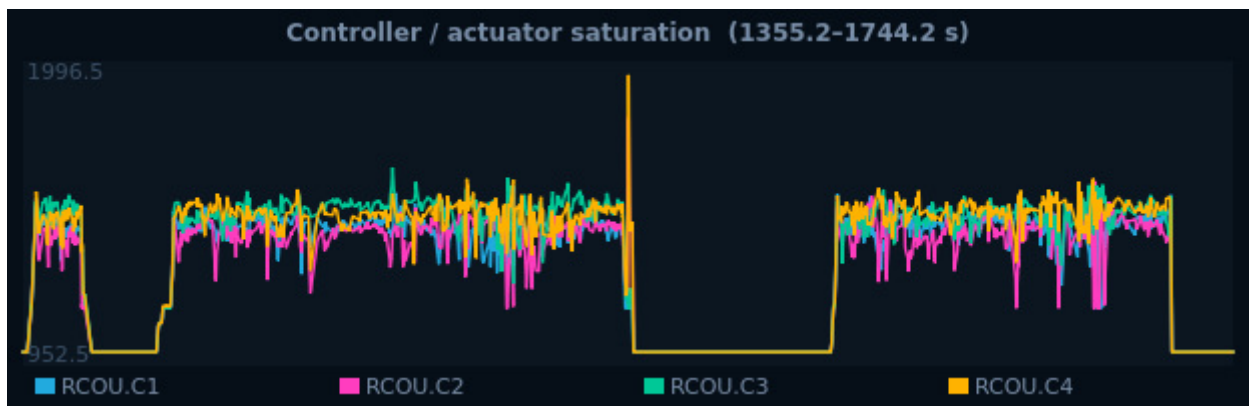
Supporting evidence: ATT.Yaw, ATT.DesYaw



6. [WARNING] SATURATION — Controller / actuator saturation

Saturated 8% of armed time (motors 8%, output 0%)

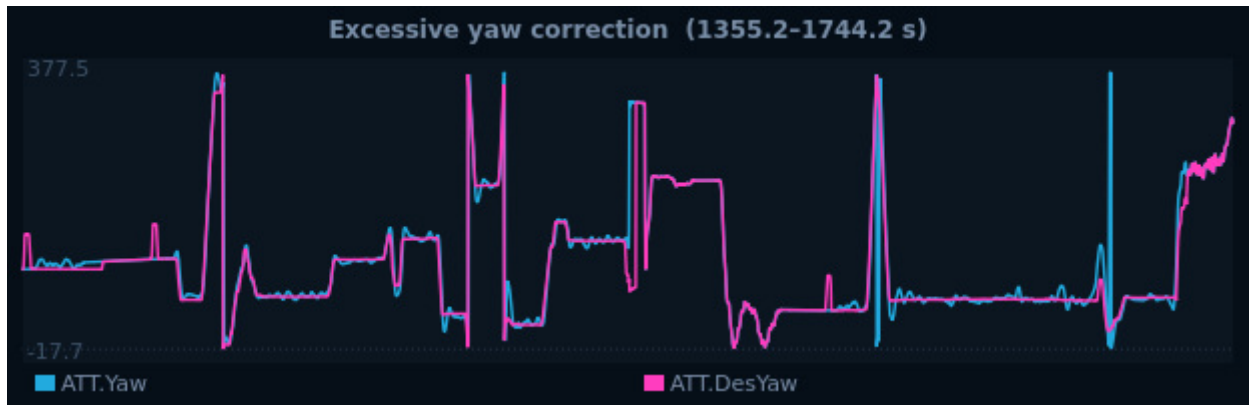
Supporting evidence: RCOU.C1-4, RATE.R0ut/P0ut/Y0ut



7. [WARNING] YAW — Excessive yaw correction

Yaw discipline score 55 (67 large yaw inputs)

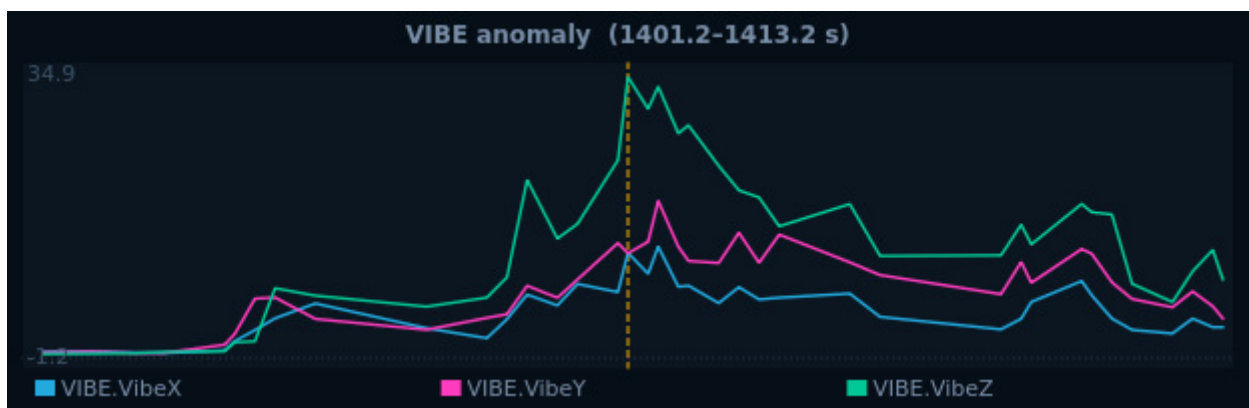
Supporting evidence: ATT.Yaw, ATT.DesYaw, RCIN(yaw)



8. [WARNING] VIBE — VIBE anomaly

High vibration VibeZ=71.9 m/s² @ 1407.2 s

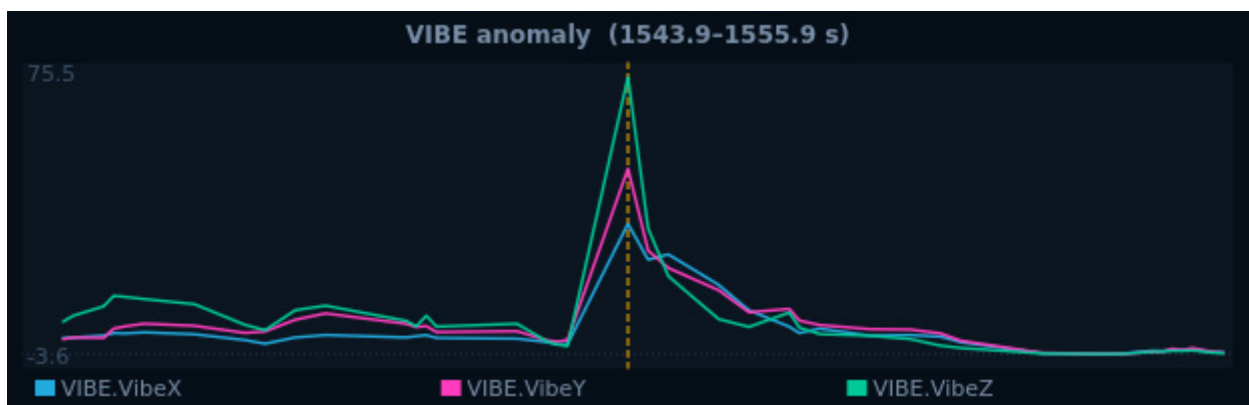
Supporting evidence: VIBE



9. [WARNING] VIBE — VIBE anomaly

High vibration VibeX=46.2 m/s² @ 1549.9 s

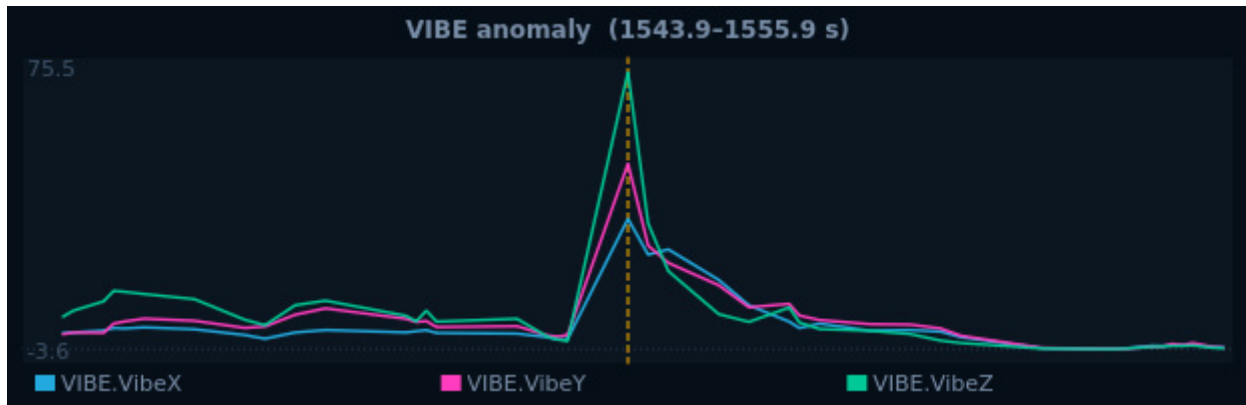
Supporting evidence: VIBE



10. [WARNING] VIBE — VIBE anomaly

High vibration VibeY=54.4 m/s² @ 1549.9 s

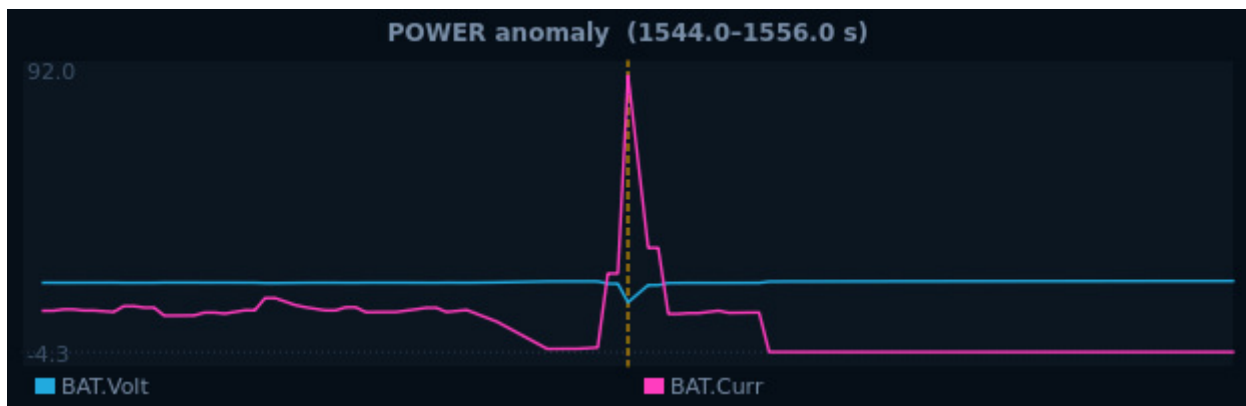
Supporting evidence: VIBE



11. [WARNING] POWER — POWER anomaly

Low battery 15.8 V @ 1550.0 s

Supporting evidence: POWER



Investigation Snapshots

Snapshot 1 — ERR: ERR: Subsys_12 code=1

Status: FLAGGED · **Captured:** 2026-06-04T12:09:31 · **Flight time:** 1552.26 s · **Flight window:** 2 / 3

Event: ERR: ERR: Subsys_12 code=1 @ 1552.26 s

Field	Value
Phase	HOVER
Mode	LOITER
Position	17.439680, 78.746050
Altitude (AGL)	-0.2 m (POS.RelHomeAlt)
Vertical speed	-0.19 m/s (BARO[0].CRt)
Ground speed	0.2 m/s
GPS	DGPS · 27 sats
EKF health	OK (ratio 0.10, SM)
Position divergence	0.14 m (OK)
Verification	NOT_LOADED

Control — Pilot / Demand / Actual:

Axis	Pilot	Demand	Actual	Δ
Roll	+0.00	-2°	-166°	164°


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yes1551.556-1552.356ekf_worstXKF4[0].SM0.1—yes1552.195-1552.295posdiv_ipnXKF3[0].IPN-0.02
—yes1551.795-1552.295posdiv_ipeXKF3[0].IPE0.1407—yes1551.795-1552.295
```

Snapshot 2 — ERR: ERR: Subsys_25 code=1

Status: FLAGGED · **Captured:** 2026-06-04T12:09:31 · **Flight time:** 1725.59 s · **Flight window:** 3 / 3

Event: ERR: ERR: Subsys_25 code=1 @ 1725.59 s

Field	Value
Phase	HOVER
Mode	LOITER
Position	17.439810, 78.746223
Altitude (AGL)	2.4 m (POS.RelHomeAlt)
Vertical speed	-0.46 m/s (BARO[0].CRT)
Ground speed	0.6 m/s
GPS	DGPS · 30 sats
EKF health	OK (ratio 0.06, SV)
Position divergence	0.06 m (OK)
Verification	NOT_LOADED

Control — Pilot / Demand / Actual:

Axis	Pilot	Demand	Actual	Δ
Roll	+0.00	-9°	+2°	11°
Pitch	+0.00	+2°	+9°	7°
Yaw	+0.00	+67°	+76°	8°
Throttle	0.00	—	0.00	—

Notes: auto from finding: ERR anomaly

[illegible]

[illegible]

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Data provenance (25 sampled values) FieldSourceValueSample t (s)Interp.Brace
(s)pilot_rollRCIN.C11501—yes1725.496-1725.696servo_rollRCOU.C11000—
yes1725.496-1725.696pilot_pitchRCIN.C21501—yes1725.496-1725.696servo_pitchRCOU.C21000—
yes1725.496-1725.696pilot_yawRCIN.C41495—yes1725.496-1725.696servo_yawRCOU.C41000—
yes1725.496-1725.696pilot_throttleRCIN.C31051—
yes1725.496-1725.696servo_throttleRCOU.C31000—
yes1725.496-1725.696demand_rollATT.DesRoll-9.314—
yes1725.496-1725.596demand_pitchATT.DesPitch1.636—
yes1725.496-1725.596demand_yawATT.DesYaw67.49—
yes1725.496-1725.596demand_throttleCTUN.ThO—no—response_rollATT.Roll2.08—
yes1725.496-1725.596response_pitchATT.Pitch8.804—
yes1725.496-1725.596response_yawATT.Yaw75.94—
yes1725.496-1725.596altitude_aglPOS.RelHomeAlt2.417—
yes1725.496-1725.696vertical_speedBARO[0].CRT-0.4597—
yes1725.496-1725.596ground_speedGPS[0].Spd0.6124—
yes1725.556-1725.956gps_statusGPS[0].Status41725.556no—gps_satsGPS[0].NSats301725.556no—
position_latGPS[0].Lat17.44—yes1725.556-1725.956position_lngGPS[0].Lng78.75—
yes1725.556-1725.956ekf_worstXKF4[0].SV0.05944—
yes1725.495-1725.595posdiv_ipnXKF3[0].IPN0.06—
yes1725.495-1725.595posdiv_ipeXKF3[0].IPE-0.01—yes1725.495-1725.595
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